Asymmetric Capital Buffer Control

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1 Introduction

1.1 Question

When a bank's capital buffer is explained by a fixed cost of issuing equity together with a penalty that grows as capital falls below the requirement, does the bank's policy carry the weight of that penalty as a parameter of its own, separate from the issuance cost, and if so where, in the dividend barrier and the recapitalisation trigger or in the variance of the buffer? The weight is never observed (Appendix C), so it can be read only off what the policy shows, and at one it switches the penalty off and returns the issuance-cost model (M8), so whether it is identified decides whether the specification Peura and Keppo [2006] call for (L18) can be tested against theirs rather than imposed (L20). The bank recapitalises by impulse control at a fixed issuance cost and takes risky exposure under a regulatory cap.

1.2 Answer

The penalty moves the thresholds. The asymmetry parameter λ_S and the fixed issuance cost K both raise the dividend barrier (M1). The recapitalisation trigger rises in λ_S and falls in K (M2). Because the two parameters move the trigger in opposite directions, the Jacobian of the barrier and the trigger in (λ_S, K) is non-singular, so the pair is locally identified from the two thresholds (M3), under the hypotheses that M4 and Appendix B list. This is K1 and C1. The thresholds identify the weight of the penalty and not whose cost it is (Appendix D). The identification is within the model. The two thresholds are recorded as the state at a dividend and at an issue, and on the panel the parent bundle read dividends are paid at every state and issues land at no single target, so neither is measured there as a threshold (Appendix C).

The penalty does not move the limits of buffer variance. Where the optimal exposure binds the cap near the distress boundary and deep in surplus, buffer variance tends to the deficit limit and the surplus limit, which involve only the asset and liability shocks and the cap (M10, M11). For every exposure that complies with the cap, binding or not, buffer variance is bounded by the same quantities (M12). The asymmetry parameter enters the thresholds and not the limits of buffer variance. This is K2 and C2, with its scope in C3. The limits need the cap to bind, and on that panel the recorded exposure sits far below the cap, so the bound of M12, which needs no binding, is the result that reaches the data (Appendix C).

Table 1: Headlines.

ID	Headline	Model rows	Scope note
K1	The asymmetry parameter λ_S and the fixed issuance cost K are locally identified from the dividend barrier and the recapitalisation trigger, because both raise the barrier and they move the trigger in opposite directions.	M1, M2, M3, M4	Local identification from the thresholds, not an estimator. Rests on the hypotheses of Appendix B, among them the verification theorem, the envelope identities, the signs of $W'(x_L)$ and N', $\Psi(x_L) > 0$ and $\rho_L > \mu'(y^*)$. Fails if the two parameters moved the trigger in the same direction (M3). It identifies the weight of the penalty, not whose cost it is (L19, Appendix D). The thresholds are recorded as the state at a dividend and at an issue, and on the panel the parent bundle read neither behaves as a threshold (Appendix C), so the identification is within the model. The fixed cost is charged per unit of liabilities (Appendix D). The responses of the thresholds to the fixed cost alone are not claimed new, since Peura and Keppo [2006] and Dai et al. [2019] tabulate them, nor is pinning the fixed cost and a second parameter from moments of the thresholds, which Peura and Keppo [2006] do by calibration (L7, L8, L9).
K2	The asymmetry parameter enters the thresholds and not the limits of buffer variance. Where the optimal exposure binds the regulatory cap near both ends, those limits depend only on the asset and liability shocks and the cap, and for every compliant exposure buffer variance is bounded by them.	M3, M10, M11, M12	That the optimum binds the cap near both ends is a hypothesis, supported only by the numerical solve of the parent bundle (Appendix B); the bound M12 needs no binding. The surplus limit needs the liquidity branch to bind deep in surplus, which needs borrowing; Bayraktar et al. [2026] take no borrowing as the main banking case, under which it never binds (L2, Appendix D). The constant buffer variance on the solvency branch is not claimed new, since it is the constant log-cushion volatility of capped portfolio insurance [Black and Perold, 1992], L5. The contrast is with that rule, whose slopes are set by preferences (L6), while here they are set by regulation and the liability shock is not hedged. On the panel the parent bundle read the cap does not bind, so only the bound of M12 reaches the data (Appendix C).

2 Model

2.1 Players

Table 2: Players of the model, the choice each faces and how the rows settle it.

ID	Player	Choice faced	How the model settles it	In the literature	Rows
P1	Bank	Its risky exposure, the dividend barrier, the recapitalisation trigger and the injection target.	The thresholds are derived, in their responses to λ_S and K (M1 to M4) and in the target's independence of the starting state (M7). The exposure is not derived. M11 takes it at the cap as a hypothesis, and M12 bounds every compliant exposure.	The bank of Peura and Keppo [2006] and of Dai et al. [2019], with a fixed issuance cost, and of Bayraktar et al. [2026], with the cap.	M1, M2, M3, M4, M7, M11, M12, L1, L7, L9
P2	Shareholders	None in the model. They receive dividends and supply each issue at the fixed and the proportional issuance cost.	A rule taken as given. The objective is their value net of the shortfall penalty, and the bank chooses for them (M8). Finite issuance follows from the fixed cost (M6).	In Peura and Keppo [2006] the owners choose the issue.	M6, M8, L7

ID	Player	Choice faced	How the model settles it	In the literature	Rows
P3	Creditors	None. The liabilities carry their own shock.	State. The liability volatility and the return-liability correlation enter the limits and bounds of buffer variance (M10 to M12).	No paper read has a liability shock unhedged by the safe asset; in Black and Perold [1992] the reserve asset tracks the liability.	M10, M11, M12, L5
P4	Supervisor	None in the model. It sets required capital and the regulatory cap.	A rule taken as given (A2). The shortfall penalty below the reference stands for its restrictions, a reading no row derives (Appendix D).	The cap of Bayraktar et al. [2026]. The intervention of Milne and Whalley [2001], which the buffer guards against (L16).	M8, M10, L1, L16

2.2 Assumptions

Table 3: Assumptions of the model.

ID	Assumption	In plain English	Symbols	Model rows	Reason
A1	The state is $x \in (1,\infty)$, with diffusion $\sigma^2(x,\pi) = \pi^2\sigma^2x^2 + 2\pi c\sigma\sigma_L x(1-x) + \sigma_L^2(1-x)^2$ under risky exposure π, and the capital buffer is $z = \log((x-1)/q)$ with $q > 0$.	The bank holds a risky and a riskless asset and owes liabilities that carry their own shock. The state is the assets-to-liabilities ratio, the risky exposure sets how much of the asset volatility the bank bears beside the liability volatility and the return-liability correlation, and the capital buffer measures equity against required capital on a log scale.	x, π, σ, σ_L, c, q, z	M9, M10, M11, M12	The balance sheet of Bayraktar et al. [2026] in ratio form. Appendix D, the balance sheet.
A2	The regulatory cap is $u(x) = \min\{(x-1)/a_1, (x-a_2)/a_3\}$ with $0 < a_1 < a_3$ and $a_2 < 1$, and the exposure complies, $0 \le \pi(x)\,x \le u(x)$.	Regulation sets a regulatory cap on the risky exposure at each assets-to-liabilities ratio, the smaller of a solvency branch, whose slope is the solvency floor and which binds near distress, and a liquidity branch, set by the run-off rate and the liquidity haircut and binding deep in surplus.	u, a_1, a_2, a_3, π, x	M10, M11, M12	The cap of Bayraktar et al. [2026]. Appendix D, the regulatory cap.
A3	Buffer variance is $\zeta(x) = z'(x)^2\,\sigma^2(x,\pi(x))$.	Buffer variance is the variance rate of the capital buffer, by the change of variable from the assets-to-liabilities ratio, at the risky exposure the bank takes and the asset volatility it bears.	ζ, z, x, π, σ	M10, M11, M12	Itô's formula, not proved in Lean. Appendix B.
A4	The objective is dividends net of issuance costs minus the running shortfall penalty $\Lambda(x) = (\lambda_S - 1)(R-x)^+$ with $\lambda_S \ge 1$, discounted at ρ_L.	Shareholders value dividends net of what they pay for issues, less a shortfall penalty for each moment the assets-to-liabilities ratio sits below the reference, weighted by the asymmetry parameter and discounted at the liability-net discount rate.	Λ, λ_S, R, ρ_L, x	M1, M2, M3, M4, M8	The asymmetric penalty is the paper's question. Appendix D, the shortfall penalty.
A5	Each issue costs K per unit of liabilities plus κ per unit raised.	Every recapitalisation pays a fixed issuance cost that does not scale with its size and a proportional issuance cost.	K, κ	M2, M5, M6, M7	The cost structure of Peura and Keppo [2006] and Bolton et al. [2011]. Appendix D, the fixed issuance cost.

ID	Assumption	In plain English	Symbols	Model rows	Reason
A6	V is the value function, solves the quasi-variational inequality with impulse operator $\mathscr{M}V$, is twice continuously differentiable on the inaction region, and the optimal policy is a dividend barrier y^*, a trigger x_L and a target y_{post}, with $\partial_{\lambda_S} V = -W$ and $\partial_K V = -N$.	The value function is the value of the bank under its best policy, which pays dividends at the dividend barrier and recapitalises from the recapitalisation trigger to the injection target, the impulse operator being the value of issuing at once, and the value responds to the asymmetry parameter and the fixed issuance cost through the expected discounted shortfall and the discounted injection count.	V, $\mathscr{M}V$, y^*, x_L, y_{post}, W, N, λ_S, K	M1, M2, M3, M4, M5	The verification theorem and the envelope identities, not proved in Lean. Appendix B.

2.3 Results

Table 4: Results of the model, each with its statement and the Lean theorems that prove it.

ID	Claim	Statement	Lean	Proof
M1	The dividend barrier rises in the asymmetry parameter and in the fixed issuance cost.	$\theta \in \{\lambda_S, K\}$, $F = W$ or N. If $V''(y^*) = 0$, $V'(y^*) = 1$, $\Lambda'(y^*) = 0$, $F'(y^*) = 0$, $\sigma^2(y^*) > 0$, $\rho_L \neq \mu'(y^*)$, $\partial_\theta V = -F$ and the chain rule holds along $y^*(\theta)$, then $dy^*/d\theta = \rho_L F(y^*)/(\rho_L - \mu'(y^*))$, positive when $0 < \rho_L$, $\mu'(y^*) < \rho_L$ and $F(y^*) > 0$.	`Identification.vppp_at_barrier`, `Identification.fpp_at_barrier`, `Identification.barrier_slope`, `Identification.barrier_slope_pos`, `Identification.control_barrier_sign_needs_gap`, `Envelope.wode_of_envelope`, `Envelope.node_of_envelope`, `Envelope.neumann_at_ystar`	p. 12
M2	The recapitalisation trigger rises in the asymmetry parameter and falls in the fixed issuance cost, the injection target rises in the fixed issuance cost, and the trigger-to-target gap widens in it.	If $V''(x)\,dx/d\theta + \partial_\theta V'(x) = 0$ along a boundary $x(\theta)$, $V''(x) \neq 0$ and $\partial_\theta V' = -F'$, then $dx/d\theta = F'(x)/V''(x)$. With $W'(x_L) > 0$, $N'(x_L), N'(y_{\text{post}}) < 0$ and $V''(x_L) > 0 > V''(y_{\text{post}})$, $dx_L/d\lambda_S > 0$, $dx_L/dK < 0$, $dy_{\text{post}}/dK > 0$ and $d(y_{\text{post}} - x_L)/dK > 0$.	`Identification.boundary_slope`, `Identification.trigger_rises_in_lambda`, `Identification.trigger_falls_in_K`, `Identification.target_rises_in_K`, `Identification.gap_increasing_in_K`, `Identification.control_trigger_slope_needs_curvature`, `Envelope.trigger_jump`	p. 12
M3	The Jacobian of the barrier and the trigger in the two parameters is non-singular, because the parameters move the trigger in opposite directions.	$\det \partial(y^*, x_L)/\partial(\lambda_S, K) = \frac{\rho_L}{(\rho_L - \mu'(y^*))V''(x_L)} \big(W(y^*)N'(x_L) - N(y^*)W'(x_L)\big)$, negative under the signs of M1 and M2. For $a_{11}, a_{12}, a_{21} > 0 > a_{22}$, $a_{11}a_{22} - a_{12}a_{21} < 0$. With $a_{22} > 0$ it can vanish.	`Identification.det_formula`, `Identification.det_neg_of_opposite`, `Identification.det_neg`, `Identification.control_same_direction_can_fail`	p. 12

ID	Claim	Statement	Lean	Proof
M4	The value function is convex just above the recapitalisation trigger, strictly so exactly when waiting is strictly suboptimal just below it.	$\Psi = \rho_L V + \Lambda - (1+\kappa)\mu$. If $\sigma^2(x_L) > 0$, $\frac{1}{2}\sigma^2 V'' + \mu V' - \rho_L V - \Lambda = 0$ at x_L^+, $V'(x_L) = 1+\kappa$ and $\mu(1+\kappa) - \rho_L V - \Lambda \le 0$ at x_L, then $\frac{1}{2}\sigma^2 V''(x_L^+) = \Psi(x_L) \ge 0$, and $V''(x_L^+) > 0 \iff \Psi(x_L) > 0$. With $\Psi(x_L) = 0$, $V''(x_L^+) = 0$.	`SecondOrder.psi_identity`, `SecondOrder.psi_nonneg_of_qvi`, `SecondOrder.trigger_convex`, `SecondOrder.trigger_strict_iff`, `SecondOrder.control_trigger_flat_when_psi_zero`	p. 13
M5	The value function has no kink at the recapitalisation trigger.	If $V \ge \mathscr{M}V$, $V(x_L) = \mathscr{M}V(x_L)$, $\mathscr{M}V$ is affine with slope $1+\kappa$, V has right derivative $V'(x_L^+)$, $\sigma^2(x_L) > 0$, and $V'(x_L^+) > 1+\kappa$ would make $\frac{1}{2}\sigma^2 q + (1+\kappa)\mu - \rho_L V - \Lambda \le 0$ hold at x_L for every q, then $V'(x_L^+) = 1+\kappa$. If $\frac{1}{2}\sigma^2 q + (1+\kappa)\mu - \rho_L V - \Lambda \le 0$ is needed only for $q \le 0$, it holds when $(1+\kappa)\mu \le \rho_L V + \Lambda$. Without $V \ge \mathscr{M}V$ a downward kink is possible.	`SmoothFit.hasDeriv_Mv`, `SmoothFit.kink_is_convex`, `SmoothFit.no_convex_kink`, `SmoothFit.smooth_fit`, `SmoothFit.control_kink_needs_obstacle`, `Envelope.smf_permits`	p. 13
M6	With a positive fixed issuance cost and a finite value, the discounted number of recapitalisations is finite, so the impulse terms of the verification argument converge absolutely.	$K > 0$, $e_i \ge 0$, $K\sum_{i<n} e_i \le B$ for all $n \Rightarrow$ $\sum_{i<n} e_i \le B/K$, $N = \sum_i e_i$ converges and $N \le B/K$. If also $\lvert c_i\rvert \le C e_i$, $\sum_i c_i$ converges absolutely. With $K = 0$, $e_i = 1$ the cost bound holds and N diverges.	`ImpulseCount.partial_sums_bounded`, `ImpulseCount.count_summable`, `ImpulseCount.count_le`, `ImpulseCount.impulse_sum_summable`, `ImpulseCount.control_count_needs_fixed_cost`	p. 13
M7	With a fixed issuance cost and a linear proportional cost, the injection target does not depend on the state from which the bank injects.	$\forall z\ \big(V(z) - (1+\kappa)z \le V(y) - (1+\kappa)y\big) \iff$ $\forall z\ \big(V(z) - (1+\kappa)(z-x) - K \le$ $V(y) - (1+\kappa)(y-x) - K\big)$, for every x. A target optimal from one x is optimal from every x'. With the convex cost $(z-x)^2$ and $V(z) = 2z$, the optimal target moves with x.	`Targeting.target_free_of_start`, `Targeting.target_same_for_all_starts`, `Targeting.control_target_moves_with_convex_cost`	p. 13
M8	At $\lambda_S = 1$ the shortfall penalty vanishes identically, and for every $\lambda_S \ge 1$ the running payoff is concave.	$\Lambda(x) = (\lambda_S - 1)(R-x)^+ \Rightarrow \Lambda \equiv 0$ at $\lambda_S = 1$, and $-\Lambda$ is concave on $\mathbb{R}$ for $\lambda_S \ge 1$. At $\lambda_S = 0$, $R = 0$ it is not concave.	`QVI_Part1.lambda_asymmetry_vanishes_at_one`, `QVI_Part1.neg_Lambda_concave`, `QVI_Part1.control_concavity_needs_lambda_ge_one`	p. 14
M9	The capital buffer maps the state space onto the real line, with required capital at zero and the distress boundary at minus infinity.	$q > 0 \Rightarrow z$ is strictly increasing on $(1,\infty)$, $1 + qe^{z(x)} = x$ for $x > 1$, $z(1+qe^w) = w$, $z(1+q) = 0$, $z(x) \to -\infty$ as $x \to 1^+$ and $z(x) \to \infty$ as $x \to \infty$.	`CapGeometry.zmap_inverse`, `CapGeometry.zmap_of_inverse`, `Coordinate.zmap_strictMonoOn`, `Coordinate.zmap_at_requirement`, `Coordinate.zmap_tendsto_atBot`, `Coordinate.zmap_tendsto_atTop`	p. 14
M10	The solvency branch of the cap binds exactly up to the crossing point $\bar{x}$, and there buffer variance is the constant deficit limit ν_1^2.	$0 < a_1 < a_3$, $a_2 < 1 \Rightarrow$ $\bar{x} - 1 = a_1(1-a_2)/(a_3 - a_1) > 0$, $u(x) = (x-1)/a_1$ for $x \le \bar{x}$, $u(x) = (x-a_2)/a_3$ for $x \ge \bar{x}$. If also $q > 0$, $1 < x \le \bar{x}$ and $\pi(x)\,x = u(x)$, then $\zeta(x) = \nu_1^2$.	`CapGeometry.hasDerivAt_zmap`, `CapGeometry.zvol2_eq`, `CapGeometry.xbar_sub_one`, `CapGeometry.one_lt_xbar`, `CapGeometry.solvency_binds`, `CapGeometry.liquidity_binds`, `CapGeometry.zvol2_solvency`	p. 14

ID	Claim	Statement	Lean	Proof
M11	For any exposure policy that binds the cap near an end of the state space, buffer variance tends to the deficit limit v_1^2 at the distress boundary and to the surplus limit v_3^2 deep in surplus.	$q > 0$, $0 < a_1 < a_3$, $a_2 < 1$. If $\pi(x)x = u(x)$ for all x near 1^+ then $\zeta(x) \to v_1^2$ as $x \to 1^+$. If $\pi(x)x = u(x)$ for all large x then $\zeta(x) \to v_3^2$ as $x \to \infty$. Both fail without $a_1 < a_3$ or without binding.	`CapGeometry.deficit_limit`, `CapGeometry.surplus_limit`, `CapGeometry.variance_ratio_is_cap_geometry`, `CapGeometry.control_surplus_limit_needs_a1_lt_a3`, `CapGeometry.control_constancy_needs_binding`	p. 14
M12	For every exposure policy that complies with the cap, binding or not, buffer variance is at most the larger of σ_L^2 and the deficit limit v_1^2 everywhere, and deep in surplus at most the larger of σ_L^2 and the surplus limit v_3^2 up to any margin.	$q > 0$, $a_1 > 0$, $x > 1$, $0 \le \pi(x)$, $\pi(x)x \le u(x) \Rightarrow \zeta(x) \le \max\{\sigma_L^2, v_1^2\}$. If $a_3 > 0$, $a_2 < 1$ and compliance holds for all large x, then for every $\varepsilon > 0$, $\zeta(x) \le \max\{\sigma_L^2, v_3^2\} + \varepsilon$ for all large x. Both fail with $\pi < 0$ or with $\pi x > u(x)$.	`Bounds.qv_le_max`, `Bounds.zvol2_eq_qv`, `Bounds.nu2_eq_qv`, `Bounds.qv_zero`, `Bounds.deficit_bound`, `Bounds.surplus_bound`, `Bounds.control_bound_needs_long_exposure`, `Bounds.control_bound_needs_compliance`	p. 15

3 Literature

3.1 How the papers read leave the question open

The papers read settle the premises of the question and leave the question itself. Banks hold capital well above the requirement (L15), and what the buffer guards against is a costly intervention by the supervisor or a response of the market once capital falls below the minimum (L16). An optimising bank with a fixed issuance cost and liquidation at the minimum underpredicts the level (L17). Peura and Keppo [2006] call that penalty extreme and ask for one that grows as the capital ratio deteriorates (L18); the shortfall penalty of A4 is that specification, with λ_S its weight. The weight must then be set, and the two papers that calibrate this class set their frictions by hand or from values. Peura and Keppo [2006] impose common values by intuition and pin the delay and the fixed cost from issue cost and frequency (L20, L8), and Dai et al. [2019] fit their parameters to equity values (L21). Neither asks whether the weight of the penalty and the fixed cost can be told apart by the bank's policy, where both raise the buffer. The weight matters because at one the penalty is off and the model is theirs (M8), so an identified weight is what would let the specification be tested rather than imposed. K1 answers that part (M1 to M3, C1, C6). Whose cost the penalty is, which Peura and Keppo [2006] name as the issue their omissions reduce to (L19), is not settled by the thresholds and stays outside the paper (Appendix D).

On the variance of the buffer the papers read point both ways. In an unconstrained loss-averse control the large-wealth limits of the controls depend on the preference parameter (L14); in an endowment economy with constant loss aversion the volatility of returns equals that of dividends (L12); in capped portfolio insurance the two volatility levels are set by the multiple and the leverage limit, which preferences choose (L5, L6). For a bank under a regulatory cap no paper read evaluates the limits. K2 does (M10 to M12, C2), and C3 says where the surplus limit applies (L2).

3.2 Papers read

Table 5: Papers read, with the passage each row relies on.

ID	Paper	What we rely on	Quote	Page	Rows
L1	Bayraktar et al. [2026]	The cap u and the crossing point $\bar{x}$ of M10 are this paper's eq. (4) and its switching point. M10 and M11 are statements about its cap. The parallel ends there, since the paper evaluates no limit of the volatility and has no asymmetry parameter.	"the parameter a3 scales the entire liquidity constraint and determines its asymptotic upper bound"	17	M10, M11
L2	Bayraktar et al. [2026]	The authors' main banking case has no borrowing. With no borrowing the liquidity branch lies above the ceiling once $a_2 \leq (1-a_3)x$, so it never binds (`BCVW.ceiling_tighter_than_lcr`), and M11's surplus limit belongs to their unrestricted benchmark.	"we treat the $\pi \leq 1$ case as the main banking interpretation and the unrestricted case as a mathematical benchmark"	19	M11
L3	Bayraktar et al. [2026]	The paper has no fixed issuance cost, fixes its recapitalisation threshold exogenously, and names a fixed cost as a change that could move recapitalisation. M2 and M3 take up that change.	"If one introduced fixed issuance costs, endogenous market access, or adverse-selection costs that rise near distress, earlier recapitalization or inaction at the boundary could become optimal."	13	M2, M3
L4	Bayraktar et al. [2026]	A fixed issuance cost charged in dollars breaks the reduction to one dimension. M1 to M3 charge K per unit of liabilities, which keeps it (Appendix D).	"Fixed issuance costs, nonlinear issuance costs, taxes, deposit insurance premia, endogenous deposit rates, balance-sheet size effects, multiple asset classes, or regulatory constraints depending on dollar levels rather than ratios would generally break the one-dimensional reduction."	9	M1, M2, M3
L5	Black and Perold [1992]	Their rule is the regulatory cap with $a_2 = 0$, multiple $1/a_1$ and leverage limit $1/a_3$ (`BlackPerold.cap_is_cppi`), and its log-cushion volatility on the multiplier branch is constant (their Prop. 2), the constancy of v_1^2 in M10. The parallel ends at the liability, which their reserve asset tracks and ours does not.	"maintains the portfolio's risk exposure a constant multiple of the excess of wealth over a floor, up to a borrowing limit"	403	M10, M11
L6	Black and Perold [1992]	In their Prop. 11 the multiple is set by the impatience to consume and the leverage limit by risk aversion, so the two volatility levels carry preference information. In M11 the slopes are regulatory and the asymmetry parameter is absent from the limits.	"The multiple depends on the utility function only through"	421	M11

ID	Paper	What we rely on	Quote	Page	Rows
L7	Peura and Keppo [2006]	The bank form of the class, with a fixed issuance cost as a fixed proportion of size, the device of M1 to M3, and each issue ending at the dividend barrier, a single target as in M7.	"As in the general case, the dividend barrier is also the level up to which the capital ratio is replenished each time a new capital issue is implemented."	2176	M1, M2, M7
L8	Peura and Keppo [2006]	They pin the issue delay and the fixed cost by calibration from issue cost and frequency, moments of the thresholds, and their Table 2 has the sign pattern of M3 in the delay and the fixed cost at small costs (`PeuraKeppo.table2_det_neg`). The parallel ends at the parameter, a delay rather than a preference, and at the method, a calibration rather than a local identification theorem.	"For D and K, we identify common estimates across banks such that their implied capital raising cost and frequency are of plausible magnitude."	2184	M3
L9	Dai et al. [2019]	Their Table 2.1 gives the responses of the dividend and recapitalisation barriers to seven parameters, the fixed cost with the signs of M1 and M2 (`DaiHuangKeppo.table21_K_signs`). They state no identification and have no shareholder asymmetry parameter, and their delay and fixed cost move the trigger the same way.	"K 0.0017 −0.0543 0 −0.0004"	76	M1, M2, M3
L10	Bolton et al. [2011]	Its authors state that the double barrier of payout and external financing is an established structure. Its fixed issuance cost is proportional to capital to keep the model homogeneous, as in M1 to M3, and each issue returns to one target, as in M7.	"While this double-barrier policy is not new (it goes back to the inventory model of Miller and Orr 1966 in corporate finance), our model provides substantial new insight"	3	M1, M7
L11	Løkka and Zervos [2005]	With proportional issuance costs only, issuance is a minimal reflection at zero, neither a lump nor a target. The fixed cost is what produces the single target of M7.	"In this problem, it is optimal to issue minimal new equity so as to reflect the reserves process at 0 and make minimal dividend payments so as to reflect the reserves process at a certain level x^* (see Section 4)."	2	M7

ID	Paper	What we rely on	Quote	Page	Rows
L12	Barberis et al. [2001]	With a constant level of loss aversion, the volatility of log returns equals that of dividend growth for every level, the parallel to the absence of λ_S from the limits of M11. The parallel ends at the mechanism, a constant price-dividend ratio in an endowment economy rather than a binding regulatory cap.	"the volatility of log returns in this model is equal to the volatility of log dividend growth, namely 12 percent."	47	M11
L13	Engle and Siriwardane [2018]	They split observed equity volatility asymmetry into a mechanical-leverage share and a market-exposure share. The parallel ends at the object, a response of variance to the sign of news rather than limits of variance across regions of a state variable.	"Volatility asymmetry is mostly explained by exposure to the aggregate market, not a mechanical leverage effect."	449	M11
L14	Li et al. [2023]	In an unconstrained loss-averse control the large-wealth limits of the optimal controls depend on the preference parameters, the contrast to M11, where a binding regulatory cap fixes them.	"our asymptotic limits differ significantly from the ones in the Merton's problem, which now sensitively depends on the reference degree parameter"	13	M11
L15	Berger et al. [2008]	Buffers well above the minimum are the fact the shortfall penalty of A4 and M8 is to explain. The parallel ends at the method, a partial-adjustment regression without a model of the thresholds.	"Since the early 1990s, however, the largest U.S. banking organizations have maintained ratios well in excess of regulatory minimums."	1	M8
L16	Milne and Whalley [2001]	The buffer guards against the supervisor's intervention or the market's response below the minimum, the reading of the penalty in Appendix D. Their cost is an audit at random times, not a graduated running penalty.	"What matters instead is the possibility of future capitalisation falling below minimum required levels and inducing a costly regulatory intervention or market response."	3	M8
L17	Peura and Keppo [2006]	The optimising bank with a fixed issuance cost (M6) and liquidation at the minimum underpredicts the capital level, on their own calibration. The shortfall penalty of A4, whose form M8 carries, is the friction added.	"The model falls somewhat short of explaining the level of capital held by banks."	2188	M6, M8
L18	Peura and Keppo [2006]	The graduated penalty they call for is Λ of A4 and M8, with λ_S its weight. The parallel ends at the call, since they do not add it.	"A model in which the penalty gradually increases as the capital ratio deteriorates would therefore be more realistic."	2192	M8

ID	Paper	What we rely on	Quote	Page	Rows
L19	Peura and Keppo [2006]	The question they leave, whose cost binds the bank's capital. The thresholds identify the weight of the penalty under any reading and do not choose among them.	"Whether these omissions matter reduces to the issue of which constituency provides the binding constraint on bank capitalization."	2192	M3, M8
L20	Peura and Keppo [2006]	Their parameters beyond the return moments are imposed, not identified from the bank's policy. M3 identifies two of them from the thresholds.	"imposed common values for other parameters (mainly guided by high level economic intuition)"	2192	M3
L21	Dai et al. [2019]	Their parameters are fitted to equity values, not identified from the thresholds.	"are calibrated by fitting the partially observed value function"	84	M3

Every paper the tables name has been read in full and has a record with Lean under `lit/`.

4 Conclusions

Table 6 holds the conclusions that rest on model rows and on papers read in full. K1 rests on M1 to M4 and gives C1 and C6, with C4 pending. K2 rests on M3 and M10 to M12 and gives C2 and C3, with C5 pending. Table 7 holds every claim that a result is new and every other claim still waiting, and each of its rows names what it waits on.

Table 6: Conclusions resting on model rows and papers read.

ID	Statement	Rows
C1	Under the hypotheses of Appendix B, the asymmetry parameter λ_S and the fixed issuance cost K are locally identified from the dividend barrier and the recapitalisation trigger. Both parameters raise the barrier, and they move the trigger in opposite directions, so the Jacobian of the two thresholds is non-singular.	M1, M2, M3, M4
C2	Wherever the optimal exposure binds the cap near both ends, the limits of buffer variance are v_1^2 and v_3^2, functions of $(\sigma, \sigma_L, c, a_1, a_3)$ only, and for every compliant exposure buffer variance is bounded by $\max\{\sigma_L^2, v_1^2\}$. The asymmetry parameter moves the thresholds (C1) and is absent from these limits and bounds. In capped portfolio insurance, whose geometry the cap shares, the slopes come from preferences, so the same limits would carry them.	M3, M10, M11, M12, L5, L6
C3	The surplus limit v_3^2 requires the liquidity branch to bind deep in surplus, which requires borrowing to hold the risky asset. With no borrowing, the main banking case of Bayraktar et al. [2026], the liquidity branch never binds once $a_2 \le (1 - a_3)x$, so the deficit limit and the bound of M12 are the results that apply there.	M11, M12, L2
C6	C1 supplies what the specification Peura and Keppo [2006] call for (L18) lacks in their calibration. The weight of the penalty and the fixed cost are set by the two thresholds (M3) rather than imposed (L20) or fitted to equity values (L21). Which constituency the penalty stands for (L19) is not settled by the thresholds (Appendix D).	M3, L18, L19, L20, L21

Table 7: Claims not yet verified, each naming what it waits on.

ID	Statement	Rows
C4	The joint local identification of C1, of a shareholder asymmetry parameter and a fixed issuance cost from the two thresholds by the signs of their responses, is new. Its numerical precedents in other parameters are Peura and Keppo [2006] and Dai et al. [2019] (L8, L9). *Pending:* the identification section of Alvarez, Le Bihan and Lippi (2014), Banque de France WP 492; Glover, Johnson, Evatt and Cheng (2023); Lambertini and Uysal (2013, 2014); each read in full with a record (`notes/novelty_check_20261023.md`, section 5).	M3, L8, L9
C5	That under a regulatory two-branch cap with an unhedged liability shock the limits of buffer variance carry no information about the asymmetry parameter, while the thresholds do (C2), is new. *Pending:* Choulli, Taksar and Zhou (2003), SIAM J. Control Optim. 41(6), read in full with a record; and a proof that the optimal exposure binds the cap near both ends, the hypothesis of M11.	M11, L5, L6

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A Proofs

Notation. The state is the assets-to-liabilities ratio $x \in (1,\infty)$, $q > 0$ required capital as a ratio, and $z(x) = \log\big((x-1)/q\big)$ the capital buffer. The risky exposure is $\pi(x)$, and the regulatory cap gives the capped exposure

$u(x)=\min\{(x-1)/a_1,(x-a_2)/a_3\}$ with crossing point $\bar{x}=(a_3-a_1a_2)/(a_3-a_1)$. The diffusion of x is $\sigma^2(x,\pi)=\pi^2\sigma^2x^2+2\pi c\sigma\sigma_L x(1-x)+\sigma_L^2(1-x)^2$, with σ the asset volatility, σ_L the liability volatility and c the return-liability correlation. Buffer variance is $\zeta(x)=z'(x)^2\,\sigma^2\big(x,\pi(x)\big)$, the exposure ratio is $\omega(x)=\pi(x)\,x/(x-1)$, and $\nu_i^2=\sigma^2/a_i^2-2c\sigma\sigma_L/a_i+\sigma_L^2$ for $i\in\{1,3\}$. V is the value function, y^* the dividend barrier, x_L the recapitalisation trigger and y_{post} the injection target. W is the expected discounted shortfall and N the discounted injection count under the optimal policy, θ is λ_S or K, and F is W or N accordingly. ρ_L is the liability-net discount rate, μ the drift of x along the cap and μ' its derivative in x, Λ the shortfall penalty and R its reference, and $\ell=(R-x)^+$ for $F=W$ and $\ell=0$ for $F=N$. Primes are derivatives in x. κ is the proportional issuance cost, $\mathscr{M}V$ the impulse operator, affine with slope $1+\kappa$ on the intervention side, and $\mu=\mu(x_L)$ in M4 and M5. In M6, e_i are the discounted impulse times, c_i the impulse terms and B the cost bound.

Proof of M1

Lean:
`Identification.vppp_at_barrier`, `Identification.fpp_at_barrier`, `Identification.barrier_slope`, `Identification.barrier_slope_pos`, `Identification.control_barrier_sign_needs_gap`, `Envelope.wode_of_envelope`, `Envelope.node_of_envelope`, `Envelope.neumann_at_ystar`.

$$\tfrac12\sigma^2V'''+\tfrac12(\sigma^2)'V''+\mu V''+\mu'V'-\rho_LV'-\Lambda'=0,\quad V''=0,\ V'=1,\ \Lambda'=0\ \Rightarrow\ \tfrac12\sigma^2V'''=\rho_L-\mu'.$$

$$\tfrac12\sigma^2F''+\mu F'-\rho_LF+\ell=0,\quad F'(y^*)=0,\ \ell(y^*)=0\ \Rightarrow\ \tfrac12\sigma^2F''=\rho_LF.$$

$$V'''\frac{dy^*}{d\theta}+\partial_\theta V''=0,\quad \partial_\theta V''=-F''\ \Rightarrow\ \frac{dy^*}{d\theta}=\frac{F''}{V'''}=\frac{\rho_LF(y^*)}{\rho_L-\mu'(y^*)}.$$

Control, $\rho_L=1$, $\mu'=2$, $F=1\Rightarrow dy^*/d\theta=-1$.

□

Proof of M2

Lean:
`Identification.boundary_slope`, `Identification.trigger_rises_in_lambda`, `Identification.trigger_falls_in_K`, `Identification.target_rises_in_K`, `Identification.gap_increasing_in_K`, `Identification.control_trigger_slope_needs_curvature`, `Envelope.trigger_jump`.

$$V''(x)\frac{dx}{d\theta}+\partial_\theta V'(x)=0,\quad \partial_\theta V'=-F'\ \Rightarrow\ \frac{dx}{d\theta}=\frac{F'(x)}{V''(x)}.$$

$$\frac{dx_L}{d\lambda_S}=\frac{W'(x_L)}{V''(x_L)}>0,\qquad \frac{dx_L}{dK}=\frac{N'(x_L)}{V''(x_L)}<0,\qquad \frac{dy_{\text{post}}}{dK}=\frac{N'(y_{\text{post}})}{V''(y_{\text{post}})}>0.$$

$$\frac{d(y_{\text{post}}-x_L)}{dK}>0,\qquad N(x_L)=N(y_{\text{post}})+1.$$

Control, $V''(x)=0$, $F'(x)=0\Rightarrow dx=0$ and $dx=1$ both solve the chain rule.

□

Proof of M3

Lean: `Identification.det_formula`, `Identification.det_neg_of_opposite`, `Identification.det_neg`, `Identification.control_same_direction_can_fail`.

$$\det\begin{pmatrix}\frac{\rho_LW(y^*)}{\rho_L-\mu'} & \frac{\rho_LN(y^*)}{\rho_L-\mu'}\\ \frac{W'(x_L)}{V''(x_L)} & \frac{N'(x_L)}{V''(x_L)}\end{pmatrix}=\frac{\rho_L}{(\rho_L-\mu')V''(x_L)}\big(W(y^*)N'(x_L)-N(y^*)W'(x_L)\big).$$

$$a_{11},a_{12},a_{21}>0>a_{22}\ \Rightarrow\ a_{11}a_{22}<0<a_{12}a_{21}\ \Rightarrow\ a_{11}a_{22}-a_{12}a_{21}<0.$$

Control, $a_{11}=a_{12}=a_{21}=a_{22}=1\Rightarrow\det=0$.

□

Proof of M4

Lean: `SecondOrder.psi_identity`, `SecondOrder.psi_nonneg_of_qvi`, `SecondOrder.trigger_convex`, `SecondOrder.trigger_strict_iff`, `SecondOrder.control_trigger_flat_when_psi_zero`.

$$\tfrac{1}{2}\sigma^2 V'' + \mu V' - \rho_L V - \Lambda = 0, \quad V' = 1+\kappa \;\Rightarrow\; \tfrac{1}{2}\sigma^2 V''(x_L^+) = \rho_L V + \Lambda - (1+\kappa)\mu = \Psi(x_L).$$

$$\mu(1+\kappa) - \rho_L V - \Lambda \le 0 \;\Rightarrow\; \Psi(x_L) \ge 0 \;\Rightarrow\; V''(x_L^+) \ge 0, \qquad \sigma^2 > 0 \;\Rightarrow\; \big(V''(x_L^+) > 0 \iff \Psi(x_L) > 0\big).$$

Control, $\sigma^2 = 1$, $V' = 1$, $\kappa = 0$, $V = \Lambda = \mu = 0$, $\rho_L = 1 \Rightarrow \Psi = 0$, $V''(x_L^+) = 0$.

□

Proof of M5

Lean: `SmoothFit.hasDeriv_Mv`, `SmoothFit.kink_is_convex`, `SmoothFit.no_convex_kink`, `SmoothFit.smooth_fit`, `SmoothFit.control_kink_needs_obstacle`, `Envelope.smf_permits`.

$$V \ge \mathscr{M}V, \;\; V(x_L) = \mathscr{M}V(x_L), \;\; (\mathscr{M}V)' = 1+\kappa \;\Rightarrow\; V'(x_L^+) \ge 1+\kappa.$$

$$V'(x_L^+) > 1+\kappa \;\Rightarrow\; \tfrac{1}{2}\sigma^2 q + (1+\kappa)\mu - \rho_L V - \Lambda \le 0 \;\; \forall q, \qquad q = \frac{2(\rho_L V + \Lambda - (1+\kappa)\mu + 1)}{\sigma^2} \;\Rightarrow\; 1 \le 0.$$

$$\Rightarrow\; V'(x_L^+) = 1+\kappa; \qquad q \le 0, \;\; (1+\kappa)\mu \le \rho_L V + \Lambda \;\Rightarrow\; \tfrac{1}{2}\sigma^2 q + (1+\kappa)\mu - \rho_L V - \Lambda \le 0.$$

Control, $v(x) = x$, $\kappa = 1$, $\mathscr{M}v(x) = 2x \Rightarrow v(0) = \mathscr{M}v(0)$, $v'(0^+) = 1 < 2$.

□

Proof of M6

Lean: `ImpulseCount.partial_sums_bounded`, `ImpulseCount.count_summable`, `ImpulseCount.count_le`, `ImpulseCount.impulse_sum_summable`, `ImpulseCount.control_count_needs_fixed_cost`.

$$K\sum_{i<n} e_i \le B, \; K > 0 \;\Rightarrow\; \sum_{i<n} e_i \le \frac{B}{K} \;\; \forall n, \qquad e_i \ge 0 \;\Rightarrow\; N = \sum_i e_i \le \frac{B}{K} < \infty.$$

$$|c_i| \le C e_i \;\Rightarrow\; \sum_i |c_i| \le CN < \infty.$$

Control, $K = 0$, $e_i = 1 \Rightarrow 0 \le 0$ for every n and $\sum_i e_i = \infty$.

□

Proof of M7

Lean: `Targeting.target_free_of_start`, `Targeting.target_same_for_all_starts`, `Targeting.control_target_moves_with_convex_cost`.

$$V(z) - (1+\kappa)(z-x) - K = \big(V(z) - (1+\kappa)z\big) + \big((1+\kappa)x - K\big),$$

$$\arg\max_z \big(V(z) - (1+\kappa)(z-x) - K\big) = \arg\max_z \big(V(z) - (1+\kappa)z\big) \quad \forall x.$$

Control, $V(z) = 2z$, cost $(z-x)^2 \Rightarrow \arg\max_z \big(2z - (z-x)^2\big) = x+1$.

□

Proof of M8

Lean: `QVI_Part1.lambda_asymmetry_vanishes_at_one`,`QVI_Part1.neg_Lambda_concave`,`QVI_Part1.control_concavity_needs_lambda_ge_one`.

$$\Lambda(x)\big|_{\lambda_S=1} = (1-1)(R-x)^+ = 0.$$

$$-\Lambda(x) = (\lambda_S - 1)\big(\min\{x,R\} - R\big), \qquad \lambda_S - 1 \geq 0,\ \ x \mapsto \min\{x,R\} \text{ concave.}$$

Control, $\lambda_S = 0$, $R = 0 \Rightarrow -\Lambda(x) = \max\{-x, 0\}$, $-\Lambda(0) = 0 < \frac{1}{2} = \frac{1}{2}(-\Lambda(-1)) + \frac{1}{2}(-\Lambda(1))$.

□

Proof of M9

Lean: `CapGeometry.zmap_inverse`,`CapGeometry.zmap_of_inverse`,`Coordinate.zmap_strictMonoOn`,`Coordinate.zmap_at_requirement`,`Coordinate.zmap_tendsto_atBot`,`Coordinate.zmap_tendsto_atTop`.

$$z(x) = \log\frac{x-1}{q}, \qquad 1 + qe^{z(x)} = x\ \ (x > 1), \qquad z(1+qe^{w}) = w.$$

$$1 < x < y \ \Rightarrow\ \frac{x-1}{q} < \frac{y-1}{q} \ \Rightarrow\ z(x) < z(y), \qquad z(1+q) = \log 1 = 0.$$

$$\frac{x-1}{q} \to 0^+\ (x \to 1^+) \ \Rightarrow\ z \to -\infty, \qquad \frac{x-1}{q} \to \infty\ (x \to \infty) \ \Rightarrow\ z \to \infty.$$

□

Proof of M10

Lean: `CapGeometry.hasDerivAt_zmap`,`CapGeometry.zvol2_eq`,`CapGeometry.xbar_sub_one`,`CapGeometry.one_lt_xbar`,`CapGeometry.solvency_binds`,`CapGeometry.liquidity_binds`,`CapGeometry.zvol2_solvency`.

$$\frac{x-1}{a_1} \leq \frac{x-a_2}{a_3} \iff (a_3-a_1)x \leq a_3 - a_1a_2 \iff x \leq \bar{x}, \qquad \bar{x} - 1 = \frac{a_1(1-a_2)}{a_3-a_1} > 0.$$

$$z'(x) = \frac{1}{x-1}, \qquad \zeta(x) = \omega(x)^2\sigma^2 - 2\,\omega(x)\,c\,\sigma\sigma_L + \sigma_L^2.$$

$$1 < x \leq \bar{x},\ \pi(x)x = u(x) \ \Rightarrow\ \omega(x) = \frac{1}{a_1} \ \Rightarrow\ \zeta(x) = v_1^2.$$

□

Proof of M11

Lean: `CapGeometry.deficit_limit`,`CapGeometry.surplus_limit`,`CapGeometry.variance_ratio_is_cap_geometry`,`CapGeometry.control_surplus_limit_needs_a1_lt_a3`,`CapGeometry.control_constancy_needs_binding`.

Near 1^+, $\bar{x} > 1$ by M10, so $1 < x < \bar{x}$ eventually and $\zeta(x) = v_1^2$ there. For $x \geq \bar{x}$,

$$\omega(x) = \frac{x-a_2}{a_3(x-1)} = \frac{1}{a_3} + \frac{1-a_2}{a_3}\,(x-1)^{-1} \to \frac{1}{a_3},$$

$$\zeta(x) = \omega(x)^2\sigma^2 - 2\,\omega(x)\,c\,\sigma\sigma_L + \sigma_L^2 \to v_3^2.$$

Controls, at $\sigma = 1$, $\sigma_L = c = 0$, $q = 1$.

$$a_1 = \tfrac{1}{2},\ a_2 = 0,\ a_3 = \tfrac{1}{4} \ \Rightarrow\ u(x) = 2(x-1),\ \zeta \equiv 4 \neq 16 = v_3^2.$$

$$\pi \equiv 0 \ \Rightarrow\ \zeta(2) = 0 \neq 4 = v_1^2.$$

□

Proof of M12

Lean: `Bounds.qv_le_max`, `Bounds.zvol2_eq_qv`, `Bounds.nu2_eq_qv`, `Bounds.qv_zero`, `Bounds.deficit_bound`, `Bounds.surplus_bound`, `Bounds.control_bound_needs_long_exposure`, `Bounds.control_bound_needs_compliance`.

$$\zeta(x) = Q(\omega), \qquad Q(\omega) = \omega^2\sigma^2 - 2\omega c\sigma\sigma_L + \sigma_L^2, \qquad \omega = \frac{\pi(x)\,x}{x-1},$$

$$Q(0) = \sigma_L^2, \qquad Q(1/a_i) = \nu_i^2.$$

$$\omega = tM,\ t \in [0,1] \ \Rightarrow\ Q(\omega) = (1-t)Q(0) + tQ(M) + \sigma^2 M^2(t^2 - t) \le \max\{Q(0), Q(M)\}.$$

$$0 \le \pi,\ \ \pi x \le u(x) \le \frac{x-1}{a_1} \ \Rightarrow\ 0 \le \omega \le \frac{1}{a_1} \ \Rightarrow\ \zeta(x) \le \max\{\sigma_L^2, \nu_1^2\}.$$

$$\pi x \le \frac{x - a_2}{a_3} \ \Rightarrow\ \omega \le w(x) = \frac{1}{a_3} + \frac{1 - a_2}{a_3(x-1)} \to \frac{1}{a_3}, \qquad Q(w(x)) \to \nu_3^2.$$

Controls, $(\sigma, \sigma_L, c) = (1,1,1)$, $\omega = -1 \Rightarrow Q = 4 > 1 = \max\{Q(0), Q(1)\}$; $(\sigma, \sigma_L, c) = (1,0,0)$, $\omega = 2 > 1 \Rightarrow Q = 4 > 1$.

□

B Evidence and verification

Every row of the model table is a theorem of Lean 4 with Mathlib. The Lean column names the theorems, and the proof of each row in Appendix A names the same ones. A theorem cited as `Namespace.name` is declared in the file `theory/lean/Namespace.lean`, a copy of the file of the same name under `lean/mathlib/`. The development is built on the toolchain `v4.32.0-rc1` against Mathlib commit `360da6f`. Every literature row rests on a record under `lit/`, whose suite checks each quotation on its page of the pinned source and builds and audits the record's Lean file. The script `theory/verify.sh` runs the suites below on every build, and `theory/VERIFICATION.md` records what they printed.

1. *The Lean build and axiom audit.* Every file under `theory/lean/` is a build root and builds. No source contains `sorry`. Every declared theorem is audited with `#print axioms`, and none depends on an axiom outside `propext`, `Classical.choice` and `Quot.sound`. Each file is identical to its source in `lean/mathlib/`.

2. *The skeleton checker.* Rules MS-1 to MS-9 of `theory/checks/verify_manuscript.py`, each also run on a well-formed sample it must accept and a faulty sample it must reject.

3. *The literature records.* The suite of every record a literature row cites.

What Lean does not carry. Buffer variance ζ is defined as $z'(x)^2\sigma^2(x, \pi(x))$, and that this is the diffusion coefficient of $z(X_t)$ is Itô's formula, which is not proved (A3). That the optimal policy binds the cap near both ends of the state space is a hypothesis of M11, which K2 and C2 state as their condition, supported only by the numerical solve of the parent bundle.

M1 to M3 take as hypotheses, in raw form, the inputs of A6. That V is the value function is the verification theorem, not proved. Differentiability of V and of the thresholds in θ is the regularity lemma and the implicit function theorem. The envelope identities $\partial_{\lambda_S} V = -W$ and $\partial_K V = -N$ are Danskin's theorem. The interior equation enters differentiated in x and in θ. The signs $W(y^*) > 0$, $N(y^*) > 0$, $N' < 0$ on the inaction region and $W'(x_L) > 0$ come from hitting-time arguments, and $W'(x_L) > 0$ holds exactly when $\rho_L W(y_{\text{post}}) < R - x_L$. $V''(x_L) > 0$ follows from M4 given $\Psi(x_L) > 0$, the strict form of the inequality the quasi-variational inequality gives on the intervention side. $V''(y_{\text{post}}) < 0$ is a hypothesis, since its derivation in the parent bundle fails (Appendix E). $\rho_L > \mu'(y^*)$ is a parameter condition. K1 and C1 state these as their hypotheses.

M5 takes the viscosity supersolution property of V at x_L as a hypothesis, in the form that a kink admits lower test functions of every curvature. M6 takes the cost bound as a hypothesis, that each impulse costs at least K and the strategy's value is finite, with $e_i = e^{-\rho_L \tau_i}$ the discounted impulse times and c_i the impulse terms of the verification argument. Interior regularity of V, C^2 on the inaction region and C^3 away from R and $\bar{x}$, is assumed wherever a row differentiates V.

Illustrations. No row rests on SymPy or on numerical sampling.

C Terminology and measurement map

The map names every object of the model once, and the manuscript uses these names and no others. The names are those of the parent bundle's `MEASUREMENT_MAP.tex`, agreed with the author on 10, 13 and 19 October 2026, and four names are proposed here, the three parameters of the cap geometry taken one by one and the value function. The objects fall into three tables. Inputs are what the model takes as given, internal objects are built from the inputs inside the model, and outputs are what the rows predict. The column Directly observed says Yes when a record of the world gives the object without any computation, which is when its last column begins with Recorded, and No otherwise. Every symbol in the table of assumptions has exactly one row here. Where a row names the panel, it is the FDIC panel of US banks that the parent bundle read, 2015 to 2024, and the figures are those of the parent map.

The map carries the reassessment of the question. The asymmetry parameter and the fixed issuance cost are never observed. The two thresholds that identify them are recorded as the state at a dividend and at an issue, and on the panel read neither behaves as a threshold. The limits of buffer variance are computed from assumed volatilities and apply where the cap binds, and on that panel it does not. What reaches the data is the bound of M12 and the two recorded thresholds as events, so K1 and K2 are results within the model, and the question is one of identification, not of measurement.

C.1 Inputs

Table 8: Inputs of the model.

Symbol	Name	Type	Represents	Directly observed	Obtained how
x	**Assets-to-liabilities ratio**	A real number above one.	Total assets over total liabilities, the state of the model (A1).	Yes	Recorded as total assets over total liabilities in the Call Report, `ASSET` over `LIAB` (parent map G1).
π	**Risky exposure**	A function of the state with values between zero and the cap over the state.	The share of assets in the risky asset (A1, A2). No row derives it; M11 takes it at the cap and M12 bounds every compliant choice.	Yes	Recorded as risk-weighted assets over total assets, `RWAJT` over `ASSET`, under risk weights of zero on the riskless and one on the risky asset (parent map G2). On the panel its median is 0.69 against a cap of 2.38 at the median state, so the cap does not bind there.
σ	**Asset volatility**	A positive real number.	The volatility of the risky asset, and the letter of the diffusion of the state (A1, A3).	No	Assumed at the baseline of Bayraktar et al. [2026], 0.08 (parent map G8). The volatility of the book ratio is computed on the panel and is 4.1 times smaller than the model's at the recorded exposure, so it is not this object.
σ_L	**Liability volatility**	A positive real number.	The volatility of the shock to liabilities (A1).	No	Assumed at the baseline of Bayraktar et al. [2026], 0.03 (parent map G9).
c	**Return-liability correlation**	A real number of absolute value below one.	The correlation between the risky return and the liability shock (A1).	No	Assumed at the baseline of Bayraktar et al. [2026], 0.20 (parent map G9).

Symbol	Name	Type	Represents	Directly observed	Obtained how
q	**Required capital**	A positive real number.	The capital regulation requires, as a ratio to liabilities, the zero of the capital buffer (A1).	No	Assumed. A floor on the leverage ratio gives a candidate, 3/97 for the 3% Tier 1 leverage ratio when Tier 1 equals book equity, and the bundle does not fix it (parent map G7). Buffer variance does not depend on it.
a_1	**Solvency floor**	A real number between zero and the liquidity haircut.	The floor on equity over risky assets, the slope of the solvency branch of the regulatory cap (A2).	Yes	Recorded in the Basel standard as the CET1 minimum of 4.5% of risk-weighted assets, the baseline of Bayraktar et al. [2026]; which tier it stands for is not fixed (parent map G4).
a_2	**Run-off rate**	A real number below one.	Net 30-day outflows as a share of liabilities, the intercept of the liquidity branch (A2).	No	Assumed at 0.05, the standard's run-off factor for stable retail deposits. The standard sets a factor per class of liability, so one rate for all liabilities is the model's reduction (parent map G5).
a_3	**Liquidity haircut**	A real number between the solvency floor and one.	The haircut on the risky asset in high-quality liquid assets, the slope of the liquidity branch (A2).	No	Assumed at 0.30, the baseline of Bayraktar et al. [2026], which is none of the standard's haircuts (parent map G6).
ρ_L	**Liability-net discount rate**	A positive real number.	The discount rate of the problem written per unit of liabilities (A4).	No	Assumed from the shareholders' discount rate, 0.12 at the baseline of Bayraktar et al. [2026] (parent map G10).
R	**Reference**	A real number above one.	The state below which the shortfall penalty applies (A4).	No	Assumed. Read as the capital level below which the supervisor's restrictions begin, which prompt corrective action sets by capital category (parent ledger A9); no row fixes it.
λ_S	**Asymmetry parameter**	A real number at least one.	How much more the objective weighs a shortfall below the reference than a surplus above it (A4). At one the shortfall penalty vanishes and the model is the issuance-cost model (M8).	No	Never observed. It enters the rows through the shortfall penalty only. M3 identifies it locally with the fixed issuance cost from the dividend barrier and the recapitalisation trigger, a statement within the model, since on the panel neither threshold is measured as one (the rows of the dividend barrier and the recapitalisation trigger below).
K	**Fixed issuance cost**	A positive real number.	The cost of an issue that does not scale with its size, per unit of liabilities (A5).	No	Never observed. Its counterpart is the part of an issue's cost that does not scale with the amount raised, which Peura and Keppo [2006] set from implied issue cost and frequency rather than record (L8). M3 identifies it locally with the asymmetry parameter from the two thresholds.
κ	**Proportional issuance cost**	A non-negative real number.	The cost of an issue per unit raised (A5).	No	Assumed at the baseline of Bayraktar et al. [2026], 0.01 for dilution and 0.02 per unit issued (parent map G11).

Symbol	Name	Type	Represents	Directly observed	Obtained how

C.2 Internal objects

Table 9: Internal objects of the model.

Symbol	Name	Type	Represents	Directly observed	Obtained how
z	**Capital buffer**	A real number, $z = \log((x-1)/q)$.	The log of equity over required capital, zero at the requirement and falling without limit towards the distress boundary (A1, M9).	No	Computed from the assets-to-liabilities ratio and required capital by its formula, whose inverse is `CapGeometry.zmap_inverse` and which is zero at the requirement, `Coordinate.zmap_at_requirement`. The parent bundle records its discrete-time form as Tier 1 capital over 8.5% of risk-weighted assets (parent map G18).
u	**Regulatory cap**	A function of the state, $u(x) = \min\{(x-1)/a_1, (x-a_2)/a_3\}$.	The largest risky exposure regulation allows at each state, the smaller of its two branches (A2).	No	Computed from the assets-to-liabilities ratio and the cap geometry by its formula. The solvency branch is the smaller up to the crossing point, `CapGeometry.solvency_binds`, and the liquidity branch beyond it, `CapGeometry.liquidity_binds`.
$\bar{x}$	**Crossing point**	A real number above one, $\bar{x}-1 = a_1(1-a_2)/(a_3-a_1)$.	The state at which the two branches of the regulatory cap are equal (M10).	No	Computed from the cap geometry by its formula, `CapGeometry.xbar_sub_one`.
Λ	**Shortfall penalty**	A function of the state, $\Lambda(x) = (\lambda_S - 1)(R-x)^+$.	The term of the objective that charges a shortfall below the reference (A4). Zero when the asymmetry parameter is one, `QVI_Part1.lambda_asymmetry_vanishes_at_one`.	No	Never observed. It is a term of the objective, and whose cost it stands for is a reading (Appendix D).
V	**Value function**	A function of the state, twice continuously differentiable on the inaction region (A6).	The value of the bank under its best policy, per unit of liabilities.	No	Never observed. Its counterpart would be the market value of equity per unit of liabilities, which the panel does not carry; Dai et al. [2019] fit their parameters to such values (L21).
$\mathscr{M}V$	**Impulse operator**	A function of the state, the supremum over targets of the value after an issue net of both issuance costs.	The value of issuing at once to the best target (A6, M5, M7).	No	A device. It exists only inside the quasi-variational inequality and the proofs of M5 and M7.

Symbol	Name	Type	Represents	Directly observed	Obtained how
W	**Expected discounted shortfall**	A function of the state, $W = -\partial_{\lambda_s} V$.	The expected discounted shortfall below the reference under the optimal policy (A6).	No	A device. It enters M1 to M3 through the envelope identity only.
N	**Discounted injection count**	A function of the state, $N = -\partial_K V$.	The expected discounted number of issues under the optimal policy (A6).	No	A device. It enters M1 to M3 through the envelope identity only; its sample counterpart, a discounted count of recorded issues, is used by no row.

C.3 Outputs

Table 10: Outputs of the model.

Symbol	Name	Type	Represents	Directly observed	Obtained how
y^*	**Dividend barrier**	A real number above one.	The state at which dividends are paid (A6). M1 gives its response to the asymmetry parameter and the fixed issuance cost.	Yes	Recorded as the state at which a dividend is paid, from dividend quarters (`EQCDIV`) and the recorded state. On the panel, 56% of bank-quarters below the solved barrier pay a dividend and payment falls as the state rises, so no barrier is measured there, and under the conservation buffer the recorded payouts reflect a rule as well as the barrier (parent map G14).
x_L	**Recapitalisation trigger**	A real number above one and below the dividend barrier.	The state at which capital is raised (A6). M2 gives its response to the two parameters.	Yes	Recorded as the state just before a recapitalisation, from net stock issuance (`EQCSXQ`) and capital from the parent holding company (`EQCBHCTR`). On the panel, issues start at a mean state of 1.11 against a solved trigger of 1.03 (parent map G15).
y_{post}	**Injection target**	A real number between the trigger and the barrier.	The state an issue reaches (A6). M7 says it does not depend on where the issue starts.	Yes	Recorded as the state just after a recapitalisation. On the panel the post-issue state rises with the pre-issue state, against the single target of M7 (parent map G15).
ζ	**Buffer variance**	A function of the state, $\zeta(x) = z'(x)^2 \sigma^2(x, \pi(x))$.	The variance per unit of time of the capital buffer (A3).	No	Computed from the capital buffer, the asset volatility, the liability volatility, the return-liability correlation and the risky exposure by its formula, `CapGeometry.zvol2_eq`. The parent bundle did not compute it, since a level needs the asset volatility, which is assumed (parent map G16).

Symbol	Name	Type	Represents	Directly observed	Obtained how
v_1^2	**Deficit limit**	A real number, $v_1^2 = \sigma^2/a_1^2 - 2c\sigma\sigma_L/a_1 + \sigma_L^2$.	The limit of buffer variance at the distress boundary where the exposure binds the solvency branch (M10, M11).	No	Computed from the asset volatility, the liability volatility, the return-liability correlation and the solvency floor by its formula, `CapGeometry.deficit_limit`. It applies where the cap binds, which on the panel it does not (the row of the risky exposure).
v_3^2	**Surplus limit**	A real number, $v_3^2 = \sigma^2/a_3^2 - 2c\sigma\sigma_L/a_3 + \sigma_L^2$.	The limit of buffer variance deep in surplus where the exposure binds the liquidity branch (M11), which needs borrowing (C3).	No	Computed from the asset volatility, the liability volatility, the return-liability correlation and the liquidity haircut by its formula, `CapGeometry.surplus_limit`. It applies in the leveraged case only (C3).

D Primitives and scope

The balance sheet. The bank holds a risky asset with volatility σ and a riskless asset, and owes liabilities with their own shock of volatility σ_L and correlation c with the risky return (A1). The riskless asset does not track the liabilities, unlike the reserve asset of Black and Perold [1992], so the liability shock enters buffer variance unhedged. Rows M9 to M12 depend on it.

The regulatory cap. The capped exposure is $u(x) = \min\{(x-1)/a_1, (x-a_2)/a_3\}$ with $0 < a_1 < a_3$ and $a_2 < 1$ (A2), the cap of Bayraktar et al. [2026]. Under these restrictions the solvency branch binds near the distress boundary and the liquidity branch deep in surplus (M10). With $a_1 \geq a_3$ the surplus limit of M11 fails, by its control. With no borrowing, $u(x) \leq x$, the liquidity branch lies above the ceiling wherever $a_2 \leq (1-a_3)x$ (`BCVW.ceiling_tighter_than_lcr`, in the record of L2), so the surplus limit of M11 is a statement about the leveraged case. Rows M10, M11, M12, K2, C2 and C3 depend on it.

The fixed issuance cost. K is charged per unit of liabilities, so the dollar cost of an issue is K times liabilities (A5). This keeps the reduction of the bank's problem to the single state x, which a fixed cost in dollars would break (L4). M1 to M3, K1 and C1 depend on it.

The shortfall penalty. The objective charges $\Lambda(x) = (\lambda_S - 1)(R-x)^+$ below the reference R (A4). The model does not say whose cost the penalty is. The reading adopted is that it stands for the supervisor's restrictions below a capital threshold, which no row derives. What λ_S identifies in K1 is the weight of that penalty, under this reading. Peura and Keppo [2006] name which constituency binds the bank's capital as the issue their omissions reduce to (L19), and Milne and Whalley [2001] name the supervisor's intervention and the market's response together (L16). The thresholds identify the weight under any of these readings and do not choose among them. K1, C1 and C6 depend on this paragraph.

The policy. That the optimal policy is a dividend barrier, a recapitalisation trigger and an injection target, and that V is the value function, is the verification theorem of the parent bundle, taken as given (A6). M1 to M5 depend on it.

E Refuted conjectures

R1, strict concavity at the injection target. The parent bundle derives $V''(y_{\text{post}}) < 0$ from the sign change of $\varphi' = V' - (1+\kappa)$ at y_{post}, positive below and negative above, and from $\varphi \in C^3$, arguing that $\varphi''(y_{\text{post}}) = 0$ would

give a zero of even order. A zero of odd order can still have $\varphi'' = 0$. With $\varphi(y) = -y^4/4$ and $y_{\text{post}} = 0$, $\varphi'(y) = -y^3$ is positive for $y < 0$ and negative for $y > 0$, while $\varphi''(0) = 0$ (`SecondOrder.refuted_target_strict`). The weak inequality survives and strictness does not, so $V''(y_{\text{post}}) < 0$ enters M2 as a hypothesis (Appendix B). C1 does not use it.

F Responses to the referee reports

Not applicable. The paper has not been through review.

G Coverage of the referee comments

Not applicable, for the same reason.